

JKBOSE 2026

CLASS 11TH

**LIMITS AND
DERIVATIVES**

MATHS

LECTURE-3

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TOPICS TO BE COVERED

1

Evaluation of Limits

2

[REDACTED]

3

Related Problems

4

EXAMPLE

Let the function $f(x)$ be defined as: $f(x) = \begin{cases} kx + 3, & \text{if } x < 2 \\ x^2, & \text{if } x \geq 2 \end{cases}$ Find the value of k such that $\lim_{x \rightarrow 2} f(x)$ exists.

(Sol) We have been given that $\lim_{x \rightarrow 2} f(x)$ exists, then

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^-} f(x)$$

$$\Rightarrow \lim_{h \rightarrow 0} f(2+h) = \lim_{h \rightarrow 0} f(2-h)$$

$$\Rightarrow \lim_{h \rightarrow 0} (2+h)^2 = \lim_{h \rightarrow 0} (k(2-h) + 3)$$

$$\Rightarrow \lim_{h \rightarrow 0} (4+h^2+4h) = \lim_{h \rightarrow 0} (2k - kh + 3)$$

$$\Rightarrow 4 + 0^2 + 4(0) = 2k - k(0) + 3$$

$$\Rightarrow 4 = 2k + 3$$

$$\Rightarrow 2k = 1$$

$$\Rightarrow k = \frac{1}{2} \text{ Hence the result.}$$

QUESTION

The value of k for which the function $f(x) = \begin{cases} k - x, & \text{if } x < 1 \\ x^2 - 1, & \text{if } x \geq 1 \end{cases}$ has a Limit at $x = 1$ is:

1 $2 \times$ (Sol) We have been given that $\lim_{x \rightarrow 1} f(x)$ exists,

2 $3 \times$ $\lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^-} f(x)$ $\Rightarrow \lim_{h \rightarrow 0} (h^2 + 2h) = \lim_{h \rightarrow 0} (k - 1 + h)$

3 $4 \times$ $\Rightarrow \lim_{h \rightarrow 0} f(1+h) = \lim_{h \rightarrow 0} f(1-h)$ $\Rightarrow 0 = k - 1 + 0$

4 $1 \checkmark$ $\Rightarrow \lim_{h \rightarrow 0} [(1+h)^2 - 1] = \lim_{h \rightarrow 0} [k - (1-h)]$ $\Rightarrow k - 1 = 0$

$\Rightarrow \lim_{h \rightarrow 0} (1 + h^2 + 2h - 1) = \lim_{h \rightarrow 0} (k - 1 + h)$ $\Rightarrow k = 1$

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 0} \frac{|x|}{x}$ is equal to

1 ☒ 1

2 ☒ -1

3 ☒ 0

4 ☒ Does not exist

$$\frac{|x|}{x} = \begin{cases} \frac{x}{x}, & x > 0 \\ -\frac{x}{x}, & x < 0 \end{cases}$$

$$\Rightarrow \frac{|x|}{x} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

$$\text{Here } \lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{h \rightarrow 0} \frac{|h|}{h} = \lim_{h \rightarrow 0} \left(\frac{h}{h} \right) = 1.$$

$$\& \lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{h \rightarrow 0} \frac{|-h|}{-h} = \lim_{h \rightarrow 0} \left(\frac{-h}{-h} \right) = -1.$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{|x|}{x} \neq \lim_{x \rightarrow 0^-} \frac{|x|}{x}$$

$\Rightarrow \lim_{x \rightarrow 0} \frac{|x|}{x}$ does not exist.

QUESTION (NCERT EXAMPLER):

$\lim_{x \rightarrow 1} [x - 1]$, where $[\cdot]$ is greatest integer function, is equal to

1 $1 \times$

2 $2 \times$

3 $0 \times$

4 ✓ Does not exists

$x = 1+h$
 (Sol) Here $\lim_{x \rightarrow 1^+} [x-1] = \lim_{h \rightarrow 0} [1+h-1]$

$\Rightarrow \lim_{x \rightarrow 1^+} [x-1] = \lim_{h \rightarrow 0} [h] = 0$

Also $x = 1-h$
 $\lim_{x \rightarrow 1^-} [x-1] = \lim_{h \rightarrow 0} [1-h-1]$

$\Rightarrow \lim_{x \rightarrow 1^-} [x-1] = \lim_{h \rightarrow 0} [-h] = -1$

$\Rightarrow \lim_{x \rightarrow 1^+} [x-1] \neq \lim_{x \rightarrow 1^-} [x-1]$

$\lim_{x \rightarrow 1} [x-1]$ does not exists.

QUESTION (NCERT EXAMPLER):

Let $f(x) = \begin{cases} x^2 - 1, & 0 < x < 2 \\ 2x + 3, & 2 \leq x < 3 \end{cases}$ the quadratic equation whose roots are and is $\lim_{x \rightarrow 2^-} f(x)$

and $\lim_{x \rightarrow 2^+} f(x)$ is

1 $x^2 - 6x + 9 = 0$

2 $x^2 - 7x + 8 = 0$

3 $x^2 - 14x + 49 = 0$

4 $x^2 - 10x + 21 = 0$

(Sol) let $a = \lim_{x \rightarrow 2^+} f(x)$ & $b = \lim_{x \rightarrow 2^-} f(x)$

$\Rightarrow a = \lim_{h \rightarrow 0} f(2+h)$ & $b = \lim_{h \rightarrow 0} f(2-h)$

$\Rightarrow a = \lim_{h \rightarrow 0} (2(2+h) + 3)$ & $b = \lim_{h \rightarrow 0} ((2-h)^2 - 1)$

$\Rightarrow a = 7, b = 3$

So, Required Quadratic equation is:

$x^2 - (a+b)x + ab = 0$

$\Rightarrow x^2 - 10x + 21 = 0$

QUESTION (NCERT EXAMPLER):

$\lim_{n \rightarrow \infty} \frac{1 + 2 + 3 + \dots + n}{n^2}, n \in \mathbb{N}$, is equal to

1 $0 \times$

2 $1 \times$

3 $1/2 \checkmark$

4 $1/4 \times$

(Sol) We have

$$\lim_{n \rightarrow \infty} \frac{1+2+3+\dots+n}{n^2} = \lim_{n \rightarrow \infty} \left(\frac{\frac{n(n+1)}{2}}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2} = \lim_{n \rightarrow \infty} \frac{(n+1)}{2n}$$

$$= \lim_{n \rightarrow \infty} \frac{n(1+\frac{1}{n})}{2n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1+\frac{1}{n}}{2} \right) = \frac{1+0}{2} = \frac{1}{2}$$

$$(a+b)^n = {}^nC_0 a^n + {}^nC_1 a^{n-1} b + \dots + {}^nC_n b^n$$

$$(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$$

QUESTION (NCERT EXAMPLER):

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} \text{ is}$$

1

n ✓

(Sol) let $y = \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} \quad \left| \frac{0}{0} \text{ form} \right.$

2

1 ✗

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{{}^n C_0 + {}^n C_1 x + {}^n C_2 x^2 + \dots + {}^n C_n x^n - 1}{x}$$

3

$-n$ ✗

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{1 + n x + {}^n C_2 x^2 + {}^n C_3 x^3 + \dots + x^n - 1}{x}$$

4

0 ✗

$$\Rightarrow y = \lim_{x \rightarrow 0} \left(\frac{n x + {}^n C_2 x^2 + {}^n C_3 x^3 + \dots + x^n}{x} \right)$$

$$\Rightarrow y = \lim_{x \rightarrow 0} \frac{n + {}^n C_2 x + {}^n C_3 x^2 + \dots + x^{n-1}}{1}$$

$$\Rightarrow y = \lim_{x \rightarrow 0} (n + {}^n C_2 x + {}^n C_3 x^2 + \dots + x^{n-1})$$

$$\Rightarrow y = n + 0$$

$$\Rightarrow \boxed{y = n}$$

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THANK
You